1 **(a)** x-coordinate 3 **[B1]**

y-coordinate 5 **[B1]**

(b) Base $AB = 4$ and height $= 3$ **[M1]**

12 **[A1]**

[Total Marks: 4]

2 **(a)** $(1, 2)$ **[B1]**

(b) Identifies the base length $PQ = 10$ **[M1]**

Sets up the area equation:

$$\frac{1}{2} \times 10 \times h = 35$$

h $= 7[M1]$

$(0, 9)$ **[A1]**

[Total Marks: 4]

3 **(a)** $A = (-3, 2)$ **[B1]**

$B = (2, -3)$ **[B1]**

(b) $(3, 2)$ **[B1]**

(c) $D = (-2, -3)$ **[B1]**

Area $= 5 \times 5 = 25$ (accept $D = (3, -3)$ if order $ACBD$ used, or strict $ABCD$ giving $D = (-3, -3)$ and Area $= 6 \times 5 = 30$) — note: for $A(-3, 2)$, $C(3, 2)$, $B(2, -3)$, $ABCD$ cannot be a rectangle; re-indexing vertices as $ACBD$ gives $D(-3, -3)$ and Area $= 6 \times 5 = 30$ **[B1]**

[Total Marks: 5]

4 **(a)** All three points A, B, and C plotted accurately on the grid **[B1]**

(b) Identifies the horizontal and vertical translation, e.g. $\vec{BA} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ OR correctly plots D on the grid **[M1]**

$(3, 5)$ **[A1]**

(c) Calculates the midpoint of AC or BD: $\left(\frac{-1+6}{2}, \frac{3+1}{2}\right)$ **[M1]**

$(2.5, 2)$ **[A1]**

[Total Marks: 5]

5 **(a)** $A = (-2, 3)$ **[B1]**

$C = (1, -2)$ **[B1]**

(b) Horizontal shift of 5 units or vertical shift of 5 units identified **[M1]**

$(-4, -2)$ **[A1]**

(c) Base = 5 and perpendicular height = 5 **[M1]**

25 **[A1]**

[Total Marks: 6]

6 (a) Uses vector shift $\overrightarrow{QR} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ applied to $P(-2, 1)$ **[M1]**

$(0, -4)$ **[A1]**

(b) $\left(\frac{-2+6}{2}, \frac{1+(-2)}{2}\right)$ **[M1]**

$(2, -0.5)$ or $\left(2, -\frac{1}{2}\right)$ **[A1]**

(c) Encloses $PQRS$ in bounding rectangle of width 8 and height 5 (40) and subtracts corner triangles: $40 - 2\left(\frac{1}{2} \times 6 \times 2\right) - 2\left(\frac{1}{2} \times 2 \times 5\right)$ **[M1]**

34 **[A1]**

[Total Marks: 6]

7 (a) All four points P, Q, R, and S plotted accurately **[B1]**

Points joined with straight line segments to form kite $PQRS$ **[B1]**

(b) $(1, 2)$ **[B1]**

(c) Diagonal lengths $PR = 9$ and $QS = 6$ **[M1]**

$\frac{1}{2} \times 9 \times 6 = 27$ **[A1]**

(d) $P' = (1, -5)$ **[B1]**

$Q' = (-4, 2)$ **[B1]**

[Total Marks: 7]

8 (a) Identifies the area of the enclosing square as a^2 and the three outer right-angled triangles as $\frac{1}{2}ab$, $\frac{1}{2}ab$, and $\frac{1}{2}(a-b)^2$ **[M1]**

Sets up the subtraction for the area of OPQ: Area $= a^2 - \frac{1}{2}ab - \frac{1}{2}ab - \frac{1}{2}(a-b)^2$ **[M1]**

Correctly expands brackets and simplifies algebraically to $\frac{1}{2}(a^2 - b^2)$ without errors **[A1]**

(b) Sets up the distance squared equation: $(a-b)^2 + (b-a)^2 = (4\sqrt{2})^2$

$2(a-b)^2 = 32$ [M1]

Solves to find $a - b = 4$ (accepting $a > b$) **[A1]**

Uses the area equation:

$$\frac{1}{2}(a^2 - b^2) = 24$$

$a^2 - b^2 = 48$ [B1]

Factorises to $(a-b)(a+b) = 48$, substitutes $a - b = 4$ to obtain $a + b = 12$ **[M1]**

$a = 8$ and $b = 4$ **[A1]**

[Total Marks: 8]

9 **(a)** Width $= x - (-2) = x + 2$ and Height $= y - (-4) = y + 4$ **[M1]**

$2(x+2) + 2(y+4) = 30 \implies 2x + 2y + 12 = 30 \implies x + y = 9$ **[A1]**

(b) Area $= (x+2)(y+4) = 54$ **[M1]**

Substitute $y = 9 - x$ to give $(x+2)(13-x) = 54$ **[M1]**

$x^2 - 11x + 28 = 0$ or equivalent three-term quadratic **[M1]**

$(4, 5)$ and $(7, 2)$ **[A1]**

(c) $(-2, 2)$ **[B1]**

$(7, -4)$ **[B1]**

[Total Marks: 8]

10 **(a)** Sets the x-coordinate of the midpoint to zero: $\dfrac{3k + k + 4}{2} = 0$ **[M1]**

-1 **[A1]**

(b) Substitutes $k = -1$ to find $A(-3, 1)$ and $B(3, 5)$ **[M1]**

$AB^2 = (3 - (-3))^2 + (5 - 1)^2 = 6^2 + 4^2 = 52$ **[M1]**

$2\sqrt{13}$ **[A1]**

(c) Lets C be $(x, 0)$ and sets up equation $AC^2 = BC^2$:
$(x+3)^2 + (0-1)^2 = (x-3)^2 + (0-5)^2$ **[M1]**

Expands both sides correctly: $x^2 + 6x + 9 + 1 = x^2 - 6x + 9 + 25$ **[M1]**

Rearranges to solve for x: $12x = 24$

x = 2[$M1$]

$(2, 0)$ **[A1]**

[Total Marks: 9]

11 **(a)** $(-6, 0)$ **[B1]**

(b) Identifies radius $OB = 6$ and angle from positive x-axis as $60°$ **[M1]**

$x = 6\cos 60° = 3$ and $y = 6\sin 60°$ $(= 3\sqrt{3})$ **[M1]**

$(3, 5.20)$ **[A1]**

(c) Calculates length $AD = 12$, and finds $AP = \dfrac{1}{4} \times 12 = 3$ **[M1]**

$(3, 0)$ **[A1]**

[Total Marks: 6]

12 **(a)** $A(0, 6)$ **[B1]**

$B(4, 0)$ **[B1]**

(b) $(4, 6)$ **[B1]**

(c) $2y + 3(2) = 12$ **[M1]**

$(2, 3)$ **[A1]**

(d) Base $OB = 4$ and height of $\triangle OPB = 3$ (the y-coordinate of P) **[M1]**

6 **[A1]**

[Total Marks: 7]

13 **(a)** Substitutes $y = 0$ to find $A(6, 0)$ **[B1]**

Substitutes $x = 0$ to find $B(0, 9)$ **[B1]**

(b) Recognises the change in x is -6 and change in y is 9 from A to B **[M1]**

Forms a valid ratio equation for either coordinate, e.g. x-coordinate $= 6 + \frac{2}{3}(-6)$ OR y-coordinate $= 0 + \frac{2}{3}(9)$ **[M1]**

$(2, 6)$ **[A1]**

(c) Substitutes $x = 4$ and $y = 3$ into the equation: $3(4) + 2(3)$ **[M1]**

Shows $12 + 6 = 18$ and clearly concludes 'Yes' **[A1]**

[Total Marks: 7]

14 **(a)** Sets $2k - 1 < 0$ and $k + 3 > 0$, giving $-3 < k < 0.5$ **[M1]**

$-2, -1, 0$ **[A1]**

(b) Substitutes coordinates of A into $y = 2x - 4$: $k + 3 = 2(2k - 1) - 4$ **[M1]**

$3k = 9 \implies k = 3$ **[A1]**

$B = (2, 4)$ **[A1]**

(c) Sets x-coordinate of midpoint to 0: $\frac{(2k - 1) + (5 - k)}{2} = 0$ or $k + 4 = 0$ **[M1]**

-4 **[A1]**

[Total Marks: 7]

15 **(a)** $h = -3$ **[B1]**

Second quadrant (accept 2nd or II) **[B1]**

(b) Width $AB = 5 - (-3) = 8$ **[M1]**

$8(k+1) = 52$ or $k + 1 = 6.5$ **[M1]**

5.5 **[A1]**

(c) $x = \dfrac{-3+5}{2} = 1$ or $y = \dfrac{-1+5.5}{2} = 2.25$ **[M1]**

$(1, 2.25)$ **[A1]**

(d) $p = 0$ **[B1]**

$\dfrac{1}{2} \times 8 \times (q+1) = 18$ or $q + 1 = 4.5$ **[M1]**

$(0, 3.5)$ **[A1]**

[Total Marks: 10]